

Student: Abdul jabbar Alhamish

Team: Purple Team Bootcamp (Golden)

Supervisor: Anmar mohammed/Mohammed Baqer

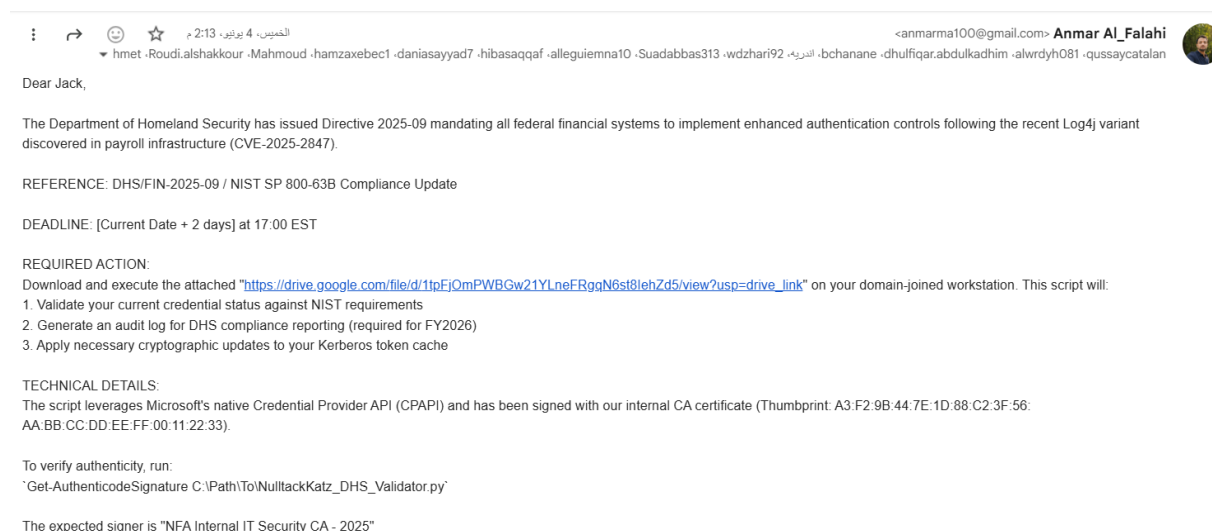
Team: purple team/Golden

Scenario receipt date: 4/6/2026

Delivery date: 22/6/2026

Scenario

On a routine Tuesday morning, a senior analyst at the National Finance Authority receives an urgent email from "IT Security." The email contains an attachment named NulltackKatz.py and threatens system lockdown if not installed immediately. The analyst, concerned about salary processing, downloads and runs the attachment. Unknown to them, the tool performs: Discovery — Maps local users, network configuration, and running processes Privilege Escalation — Enables SeDebugPrivilege for LSASS access Credential Dumping — Executes Mimikatz to extract passwords and NTLM hashes Exfiltration — Sends stolen credentials to an external Gmail account



we conducted a Purple Team exercise that combined both offensive and defensive operations ,Today The scenario began with a phishing email sent to our .defensive operations crafted to appear legitimate and urging users to enable enhanced ,environment allowing the ,This email served as the initial access vector .authentication controls Red Team to simulate a realistic social-engineering attack while the Blue Team and responded to the threat. ,detected ,analyzed

Contents

3-----	Executive Summary
4-----	Sysmon Defensive Setup
5-----	Sysmon Execution Status
6-----	Initial Attack Phase
6-----	Transition to Analysis Phase
6-----	Sysmon Event ID 10 — LSASS Access Detection
7-----	Process Tree Analysis — Sysmon Event ID 1
8-----	Network Traffic Analysis — T1041 Exfiltration Detection
8-----	Volatile Memory Investigation — Detecting Credential Theft in RAM
9-----	Memory Forensics Analysis Using Volatility
14-----	System Information — windows.info
15-----	Does the memory dump show the executed commands?
15-----	Group Identity & Attribution
15-----	Historical Operations
16-----	Technical TTP Analysis
Infrastructure & Indicators of Compromise (IoCs)-----	17
Domains Used by APT28 (with Purpose) -----	17
IP Addresses Attributed to APT28-----	17
Knowledge Graph&attack Workflow -----	20
Lessons Learned-----	21
Conclusion-----	21

Executive Summary

a Russian GRU-linked cyber , (This report analyzes APT28 (Fancy Bear The group targets governments and .espionage group active since 2004 and ,zero-day exploits ,critical infrastructure using spear-phishing the ,Based on the findings .custom malware like XAgent and CredoMap isolating ,report recommends implementing Zero Trust Architecture and deploying DNS security filtering to defend ,management interfaces against APT28-style attacks.

Sysmon Defensive Setup

Since we will be executing both the offensive and defensive components of the Purple Team exercise, we begin by preparing the defensive controls before launching the attack simulation. Our first step is configuring Sysmon, a detailed system-level monitoring tool that records critical Windows events and forwards them to the Event Log for analysis. For this exercise, we selected four essential Sysmon rules that provide high-value visibility into process behavior, network activity, and potential credential-access attempts:

- Event ID 1 – Process Creation: Logs every process executed on the system, including command-line arguments and parent-child relationships. This is crucial for detecting suspicious or unexpected process activity.
- Event ID 3 – Network Connections: Records outbound network connections initiated by processes. This helps identify unusual communication patterns, potential command-and-control traffic, or data exfiltration attempts.
- Event ID 10 – Process Access: Triggers when a process attempts to access the memory of another process. This is a key indicator of credential-dumping behavior, such as attempts to read LSASS memory.
- Event ID 7 – Image Loaded: Logs when a process loads a DLL or executable image into memory. This provides visibility into malicious DLL injection, unauthorized module loading, or execution of untrusted binaries.

These Sysmon rules establish the defensive telemetry baseline, enabling the Blue Team to detect and analyze malicious activity once the Red Team initiates the attack chain.

```
<?xml version="1.0"?>
<Sysmon schemaversion="4.90">
  <EventFiltering>
    <RuleGroup name="Process_Create_Execution" groupRelation="or">
      <ProcessCreate onmatch="include">
        <Rule name="NulltackKatz Execution">
          <Image condition="contains">NulltackKatz.py</Image>
        </Rule>
        <Rule name="Mimikatz Execution">
          <Image condition="contains">mimikatz.exe</Image>
        </Rule>
        <Rule name="PowerShell Encoded Command">
          <Image condition="is">powershell.exe</Image>
          <CommandLine condition="contains">-enc</CommandLine>
        </Rule>
      </ProcessCreate>
    </RuleGroup>
    <RuleGroup name="LSASS_Detection" groupRelation="or">
      <ProcessAccess onmatch="include">
        <Rule name="LSASS Memory Access">
          <TargetImage condition="is">lsass.exe</TargetImage>
        </Rule>
      </ProcessAccess>
    </RuleGroup>
    <RuleGroup name="Suspicious_DLL_Load_Detection" groupRelation="or">
      <ImageLoad onmatch="include">
        <Rule name="Mimikatz DLL Load">
          <ImageLoaded condition="contains">mimikatz</ImageLoaded>
        </Rule>
        <Rule name="PowerShell DLL Load">
          <Image condition="is">powershell.exe</Image>
        </Rule>
      </ImageLoad>
    </RuleGroup>
    <RuleGroup name="SMTP_Exfiltration_Detection" groupRelation="or">
      <NetworkConnect onmatch="include">
        <Rule name="SMTP 587">
          <DestinationPort condition="is">587</DestinationPort>
        </Rule>
        <Rule name="SMTPS 465">
          <DestinationPort condition="is">465</DestinationPort>
        </Rule>
      </NetworkConnect>
    </RuleGroup>
  </EventFiltering>
</Sysmon>
```

We began by implementing the selected Sysmon rules to activate them within the defensive environment. These rules form the core telemetry required for monitoring process behavior, network activity, and potential malicious actions inside Windows. The configuration includes four key Event IDs

Sysmon Execution Status

```
PS C:\NulltackLab> cd C:\NulltackLab
>>
PS C:\NulltackLab> dir

Directory: C:\NulltackLab

Mode                LastWriteTime         Length Name
----                -
d-----          6/7/2026   5:57 PM                evidence
d-----          6/8/2026   7:06 PM                logs
-a-----          6/6/2026  11:31 PM             1724 A - Copy.xml
-a-----          6/8/2026   6:59 PM      2147483648 memory.raw
-a-----          6/8/2026   6:59 PM       527640 winpmem.exe

PS C:\NulltackLab> explorer C:\NulltackLab
>>
PS C:\NulltackLab> Get-Service Sysmon

Status  Name      DisplayName
-----
Running Sysmon    Sysmon

PS C:\NulltackLab>
```

After creating the required directories (NulltackLab, logs, and evidence), we proceeded to start Sysmon to activate the monitoring rules that will capture all relevant system events before initiating the attack phase. At

the same time, we launched Wireshark to begin capturing network traffic, ensuring that both host-level and network-level telemetry were active prior to executing the offensive steps.

Initial Attack Phase

```
=====
PHISHING EMAIL CONTENT
=====
From: IT Security <security@payroll-update.com>
Reply-To: it-security@payroll-update.com
To: Finance Department <finance@desktop-211q89.local>
Subject: URGENT: Payroll System Security Update Required

Dear Finance Team,

Microsoft has released a critical security update for the Payroll
Verification System. Failure to install this update by EOD will
result in system lockout and delayed salary processing.

WARNING: Download and run the attached "NulltackKatz.py"
immediately to verify your credentials.

Thank you,
IT Security Team

[Attachment: NulltackKatz.py]
=====
OK! Phishing email saved to: C:\NulltackLab\logs\phishing_email_BLUEDELTA_20260621_160021.txt
! Student task: Analyze this email for red flags
=====
PHASE 2] T1204.002 - Malicious File Execution
=====
* I Simulating user execution of NulltackKatz.py...
  - User double-clicks the attachment
  - UAC prompt appears (if configured)
  - User clicks "yes" to run
  - Execution time: 2020-06-21T16:02:41.516265
```

Q1: What is the sender's display name?

A1: IT Security <security@payroll-update.com>

A2: What is the actual reply-to address?

Q2: : itsecurity@urgentpayroll.com

A3: What urgent claim is used?

Q3: A critical Microsoft security update for the Payroll System must be installed immediately

Q4: What MITRE technique ID covers this attack vector?

A4: T1566.001 — Spearphishing Attachment

Transition to Analysis Phase

After successfully completing Task 3, it is now appropriate to move into the detailed analysis phase. This stage focuses on reviewing all collected outputs, identifying patterns, and extracting relevant Indicators of Compromise (IOCs) associated with the malicious activity.

Sysmon Event ID 10 — LSASS Access Detection

Q1: What is the SourceImage that accessed lsass.exe?

A1: SourceImage: C:\ProgramData\Microsoft\Windows Defender\Platform\4.18.26050.15-0\MsMpEng.exe

Q2: What is the GrantedAccess value?

A2: 0x1400

Q3: What does 0x1400 represent?

A3: Query Information + Query Limited Information

Q4: What is the TargetImage?

A4: C:\Windows\System32\lsass.exe

Q5: Why is this event the PRIMARY IoC for T1003.001?

A5: Because it logs any attempt to access LSASS, which is the core behavior of Credential Dumping (T1003.001)

Process Tree Analysis — Sysmon Event ID 1

```
ParentCommandLine: "C:\Users\budhn\AppData\Local\Microsoft\WindowsApps\python.exe" NulltackKatz.py
ParentUser: DESKTOP-2L1IG9\budhn
TimeCreated : 6/18/2026 3:50:25 PM
ProviderName : Microsoft-Windows-Sysmon
Id : 1
Message : Process Create:
          RuleName:
          UtcTime: 2026-06-18 12:50:25.213
          ProcessGuid: {396a7a3b-e791-6a33-0803-000000000f00}
          ProcessId: 804
          Image: C:\Program Files\WindowsApps\PythonSoftwareFoundation.PythonManager_26.2.240.0_x64__3847v3x7pw1kn\python.exe
          FileVersion: 26.2
          Description: Python Install Manager
          Product: manage
          Company: Python Software Foundation
          OriginalFileName: python.exe
          CommandLine: "C:\Users\budhn\AppData\Local\Microsoft\WindowsApps\python.exe" NulltackKatz.py
          CurrentDirectory: C:\NulltackLab\
          User: DESKTOP-2L1IG9\budhn
          LogonGuid: {396a7a3b-e3bf-6a33-e27e-100000000000}
          LogonId: 0x107EE2
          TerminalSessionId: 1
          IntegrityLevel: High
          Hashes: SHA256=D815EE6D04C46D0699FE62C268000BAC29083EF7EF65C225022667806AC1C
          ParentProcessGuid: {396a7a3b-e63f-6a33-d802-000000000f00}
          ParentProcessId: 7612
          ParentImage: C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe
          ParentCommandLine: "C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe"
          ParentUser: DESKTOP-2L1IG9\budhn
```

Q1: What was the parent process of NulltackKatz?

A1: ParentImage:

C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe

Q2: What command-line arguments were passed?

A2: CommandLine:

"C:\Users\bwdhn\AppData\Local\Microsoft\WindowsApps\python.exe"
NulltackKatz.py

Q3: What user context did it run under?

A3: User: DESKTOP-2L1IG9\bwdhn

With the process tree successfully analyzed and all required details extracted—including the parent process, command-line arguments, and user context—the investigation can now progress to the next stage. This transition allows for correlating process-level activity with other collected evidence, strengthening the overall understanding of the attack chain and supporting deeper forensic analysis

Network Traffic Analysis — T1041 Exfiltration Detection

No.	Time	Source	Destination	Protocol	Length	Info
78	53.867584400	192.168.174.128	108.177.119.108	TLSv1.3	118	Application Data
80	53.652588400	108.177.119.108	192.168.174.128	TLSv1.3	840	Application Data, Application Data
81	53.654379400	192.168.174.128	108.177.119.108	TLSv1.3	141	Application Data
83	54.060202300	108.177.119.108	192.168.174.128	TLSv1.3	96	Application Data
84	54.060628400	192.168.174.128	108.177.119.108	TLSv1.3	82	Application Data
86	54.267181300	108.177.119.108	192.168.174.128	TLSv1.3	154	Application Data
96	54.675234100	192.168.174.128	20.233.83.145	TLSv1.3	571	Client Hello (SNI=github.com)
98	54.779414700	20.233.83.145	192.168.174.128	TLSv1.3	1454	Server Hello, Change Cipher Spec, Application Data
101	54.782670700	20.233.83.145	192.168.174.128	TLSv1.3	342	Application Data, Application Data, Application Data
102	54.785990600	192.168.174.128	20.233.83.145	TLSv1.3	118	Change Cipher Spec, Application Data
104	54.787003000	192.168.174.128	20.233.83.145	TLSv1.3	227	Application Data
106	55.085512500	20.233.83.145	192.168.174.128	TLSv1.3	133	Application Data
107	55.085516300	20.233.83.145	192.168.174.128	TLSv1.3	133	Application Data
111	55.702180600	20.233.83.145	192.168.174.128	TLSv1.3	1514	Application Data
112	55.702180700	20.233.83.145	192.168.174.128	TLSv1.3	1394	Application Data
114	55.703335900	20.233.83.145	192.168.174.128	TLSv1.3	1454	Application Data
115	55.704012300	20.233.83.145	192.168.174.128	TLSv1.3	238	Application Data, Application Data
123	56.008058000	192.168.174.128	105.199.110.133	TLSv1.3	571	Client Hello (SNI=raw.githubusercontent.com)
130	56.244090500	105.199.110.133	192.168.174.128	TLSv1.3	1454	Server Hello, Change Cipher Spec, Application Data

> Frame 86: Packet, 154 bytes on wire (1232 bits), 154 bytes captured (1232 bits) on interface \Device\NPF_{540CC94E-A36-4302-8000-000000000000} (00:00:00:00:00:00:00:00)	0000 00 0c 29 60 4c 23 00 50 56 f8 77 ac 08 00 45 00 ...}L:P V:w...E:
> Ethernet II, Src: VMware_f8:77:ac (00:50:56:f8:77:ac), Dst: VMware_60:4c:23 (00:0c:29:60:4c:23)	0010 00 0c ad ea 00 00 00 06 39 3b 6c b1 77 6c c0 a88;lwl...
> Internet Protocol Version 4, Src: 108.177.119.108, Dst: 192.168.174.128	0020 ae 00 02 4b c6 ca 67 b2 33 6c 9c 76 0d a7 50 19 ...K:g 3l:v:P:
> Transmission Control Protocol, Src Port: 587, Dst Port: 50890, Seq: 5235, Ack: 843, Len: 100	0030 fa f0 61 c9 00 00 17 03 03 00 5f 4e 04 f0 7d 2f ...a...N-)/
> Transport Layer Security	0040 fc 5b c5 97 05 4b 0c 34 21 2b 40 cd 7e 21 09 24 ...K:4 Hg-w:1:\$
	0050 17 0e a7 c7 57 12 47 b4 54 ea 00 e0 41 1d 67 3b ...W:6 T-A-B:
	0060 a3 cf d5 33 9a d5 00 1e 42 c1 37 ed ed 69 0a bc ...3...8-7-1...
	0070 8e 3d f9 fu a4 2f 4b 2d 58 70 da f8 ff a4 ff 38 .../K-Xp...8
	0080 b9 7f 5f ff ed 3b 33 40 cd e9 f6 d1 1f 32 07 53 ...:302:S
	0090 7b c1 d9 a9 70 a3 60 5c 80 e0 {...x'\...

Q1: What destination IP range received the exfiltration?

A1: 108.177.119.108 and 20.233.83.145

Q2: What destination port was used?

A2: Port 443

Q3: Is the traffic encrypted?

A3: Protocol = TLSv1.3

Info = Application Data (encrypted payload)

Q4: How could a SOC detect this without decrypting?

A4: Client Hello showed SNI = github.com

BUT the actual IP = 20.233.83.145 (Azure)

Volatile Memory Investigation — Detecting Credential Theft in RAM

```
PS C:\Users\bwdhdm\Downloads\x64> dir
Directory: C:\Users\bwdhdm\Downloads\x64

Mode                LastWriteTime         Length Name
----                -
-a-----         1/18/2023 12:48 AM        155784 Bin2Dmp.exe
-a-----         1/18/2023 12:49 AM        147784 Dmp2Bin.exe
-a-----         1/18/2023 12:49 AM        138880 Dmp2Json.exe
-a-----         1/18/2023 12:49 AM        638600 DumpIt.exe
-a-----         1/18/2023 12:49 AM        27864 Hipe2Bin.exe
-a-----         1/18/2023 12:49 AM        167240 Hipe2Dmp.exe
-a-----         1/18/2023 12:49 AM        136520 Z2Dmp.exe

PS C:\Users\bwdhdm\Downloads\x64> .\DumpIt.exe
DumpIt 3.6.20230117 (x64) (Jan 17 2023)
Copyright (C) 2007 - 2021, Matt Suiche (msuiche)
Copyright (C) 2016 - 2021, Comae Technologies DMCC (https://www.comae.com)
Copyright (C) 2022, Magnet Forensics, Inc. (https://www.magnetforensics.com/)
All rights reserved.

Thanks for using DumpIt! Always use Microsoft crash dumpst
Destination path: C:\Users\bwdhdm\Downloads\x64\DESKTOP-2LIIG09-20260621-151308.dmp
Computer name: DESKTOP-2LIIG09

--> Proceed with the acquisition ? [y/n] y

[+] Information:
Dump Type: Microsoft Crash Dump

[+] Machine Information:
Windows version: 10.0.19045
MachineId: 80C34B5E-851F-FF80-FF8D-B0B18D604C23
TimeStamp: 134263233907023260
CPU: 0x130002
MCPDataBlock: 0xfffff80460f2f680
MCPDataBlock: 0xfffff80461c1f160
MCPDataBlock: 0xfffff80461c1f160
Current date/time: [2026-06-21 (YYYY-MM-DD) 15:13:10 (UTC)]
Progress: 100.00%
```

A RAM dump was successfully captured from the target machine using DumpIt v3.6 tool. The memory image was saved as a Microsoft Crash Dump format for further analysis with Volatility 3.

Dump Details:

- Tool Used : DumpIt v3.6.20230117
- Dump Type : Microsoft Crash Dump
- Computer Name : DESKTOP-2LIIG09
- Windows Ver : 10.0.19045
- Capture Time : 2026-06-21 15:13:10 (UTC)
- File Size : 2047 MB
- Output Path : C:\Users\bwdhdm\Downloads\x64\DESKTOP-2LIIG09-20260621-151308.dmp

Memory Forensics Analysis Using Volatility

```
PS C:\Users\bwdhdm\Downloads\volatility3-develop> python3 vol.py -f "C:\Users\bwdhdm\Downloads\x64\DESKTOP-2LIIG09-20260621-151308.dmp" windows.info
Volatility 3 Framework 2.28.1
Progress: 100.00% PDB scanning finished
Variable Value
-----
Kernel Base 0xf80460a1f000
DTB 0x1ad000
Symbols file C:\Users\bwdhdm\Downloads\volatility3-develop\volatility3\symbols\windows\ntkrnlmp.pdb\F57E740B888E5056E8AF0772F1C05BEB-1.json.xz
Is64Bit True
IsPE False
layer_name 0 WindowsIntel132e
memory_layer 1 WindowsCrashDump64Layer
base_layer 2 FileLayer
MCPDataBlock 0xf8046162e400
Major/Minor 15.49041
MachineType 34404
KeNumberProcessors 2
SystemTime 2026-06-21 15:13:10+00:00
NtSystemRoot C:\Windows
NtProductType NtProductWinNt
NtMajorVersion 10
NtMinorVersion 0
PE MajorOperatingSystemVersion 10
PE MinorOperatingSystemVersion 0
PE Machine 34404
PE TimeDateStamp Sat Feb 2 23:04:03 1985
PS C:\Users\bwdhdm\Downloads\volatility3-develop>
```

Q1: What is the OS version identified from the memory image?

A1: Windows 10 — Version 10.0.19041 (64-bit)

Q2: What is the system capture time?

A2: 2026-06-21 15:13:10 UTC

Q3: What memory dump format was used?

A3: WindowsCrashDump64Layer — captured using DumpIt v3.6

Q4: How many processors does the system have?

A4: 2 Processors

Q5: What does windows.info confirm about the investigation?

A5: It confirms the memory image is valid and belongs to a 64-bit Windows 10 machine, providing the baseline system information needed before running deeper forensic analysis.

```
PE TimeDateStamp Sat Feb 2 23:04:03 1985
PS C:\Users\budhm\Downloads\volatility3-devel> python3 vol.py -f "C:\Users\budhm\Downloads\x64-DESKTOP-2L11G09-20260621-151308.dmp" windows.pslist
Volatility 3 Framework 2.20.1
Progress: 100.00
PDB scanning finished

PID PPID ImageFileName PDB Offset(U) Threads Handles SessionId Wow64 CreateTime ExitTime File output
4 0 System 0xc90538a00000 164 - N/A False 2026-06-18 12:22:21.000000 UTC N/A Disabled
32 4 Registry 0xc90538ae9000 4 - N/A False 2026-06-18 12:22:15.000000 UTC N/A Disabled
328 4 smss.exe 0xc9053927a000 2 - N/A False 2026-06-18 12:22:21.000000 UTC N/A Disabled
444 432 csrss.exe 0xc90539cb9140 10 - 0 False 2026-06-18 12:22:24.000000 UTC N/A Disabled
524 432 wininit.exe 0xc90539cc140 1 - 0 False 2026-06-18 12:22:31.000000 UTC N/A Disabled
532 516 csrss.exe 0xc905392100c0 12 - 1 False 2026-06-18 12:22:31.000000 UTC N/A Disabled
592 516 winlogon.exe 0xc90539bb4000 3 - 1 False 2026-06-18 12:22:31.000000 UTC N/A Disabled
660 524 services.exe 0xc9053d01d000 5 - 0 False 2026-06-18 12:22:31.000000 UTC N/A Disabled
684 524 lsass.exe 0xc9053d01f000 8 - 0 False 2026-06-18 12:22:31.000000 UTC N/A Disabled
788 660 svchost.exe 0xc9053d0cf240 14 - 0 False 2026-06-18 12:22:31.000000 UTC N/A Disabled
812 524 fontdrvhost.exe 0xc9053d0e4140 5 - 0 False 2026-06-18 12:22:32.000000 UTC N/A Disabled
868 660 svchost.exe 0xc9053d0e2e0 8 - 0 False 2026-06-18 12:22:32.000000 UTC N/A Disabled
944 592 fontdrvhost.exe 0xc9053d195140 5 - 1 False 2026-06-18 12:22:33.000000 UTC N/A Disabled
1020 660 svchost.exe 0xc9053d1ee2e0 15 - 0 False 2026-06-18 12:22:33.000000 UTC N/A Disabled
1376 660 svchost.exe 0xc9053d22e0c0 20 - 0 False 2026-06-18 12:22:33.000000 UTC N/A Disabled
384 592 dm.exe 0xc9053d24f000 15 - 1 False 2026-06-18 12:22:33.000000 UTC N/A Disabled
724 660 svchost.exe 0xc9053d2532e0 59 - 0 False 2026-06-18 12:22:33.000000 UTC N/A Disabled
1080 660 svchost.exe 0xc9053d2d72e0 20 - 0 False 2026-06-18 12:22:34.000000 UTC N/A Disabled
1232 660 svchost.exe 0xc9053d38a2e0 17 - 0 False 2026-06-18 12:22:34.000000 UTC N/A Disabled
1344 4 MemCompression 0xc9053d3be040 66 - N/A False 2026-06-18 12:22:34.000000 UTC N/A Disabled
1412 660 svchost.exe 0xc9053d3be2e0 17 - 0 False 2026-06-18 12:22:34.000000 UTC N/A Disabled
1516 660 svchost.exe 0xc9053d420000 8 - 0 False 2026-06-18 12:22:34.000000 UTC N/A Disabled
1572 660 svchost.exe 0xc9053d480000 3 - 0 False 2026-06-18 12:22:34.000000 UTC N/A Disabled
1592 660 svchost.exe 0xc9053d47f000 4 - 0 False 2026-06-18 12:22:34.000000 UTC N/A Disabled
1756 660 spoolsv.exe 0xc9053d582200 7 - 0 False 2026-06-18 12:22:34.000000 UTC N/A Disabled
1836 660 svchost.exe 0xc9053d5b2240 3 - 0 False 2026-06-18 12:22:34.000000 UTC N/A Disabled
1904 660 svchost.exe 0xc9053d5b52e0 12 - 0 False 2026-06-18 12:22:34.000000 UTC N/A Disabled
1944 660 svchost.exe 0xc9053d62240 9 - 0 False 2026-06-18 12:22:35.000000 UTC N/A Disabled
2128 660 svchost.exe 0xc9053d68b2e0 5 - 0 False 2026-06-18 12:22:35.000000 UTC N/A Disabled
2136 660 MpDefenderCore 0xc9053d68a000 11 - 0 False 2026-06-18 12:22:35.000000 UTC N/A Disabled
2196 660 Sysmon.exe 0xc9053d771200 13 - 0 False 2026-06-18 12:22:35.000000 UTC N/A Disabled
2252 660 vn3dservice.exe 0xc9053d7fa2e0 3 - 0 False 2026-06-18 12:22:35.000000 UTC N/A Disabled
2264 660 UGAuthService.exe 0xc9053d7fc300 2 - 0 False 2026-06-18 12:22:35.000000 UTC N/A Disabled
2280 660 Velociraptor.exe 0xc9053d8020c0 10 - 0 False 2026-06-18 12:22:35.000000 UTC N/A Disabled
2288 660 vntoolsd.exe 0xc9053d803000 12 - 0 False 2026-06-18 12:22:35.000000 UTC N/A Disabled
2296 660 MsMpEng.exe 0xc9053d808340 10 - 0 False 2026-06-18 12:22:35.000000 UTC N/A Disabled
2424 660 svchost.exe 0xc9053d86f000 11 - 0 False 2026-06-18 12:22:36.000000 UTC N/A Disabled
2556 2252 vn3dservice.exe 0xc9053d9392e0 4 - 1 False 2026-06-18 12:22:36.000000 UTC N/A Disabled
2104 660 dllhost.exe 0xc9053dc24200 10 - 0 False 2026-06-18 12:22:37.000000 UTC N/A Disabled
3088 788 dllhost.exe 0xc9053dc402c0 4 - 0 False 2026-06-18 12:22:37.000000 UTC N/A Disabled
3100 788 unsecapp.exe 0xc9053dc3a000 3 - 0 False 2026-06-18 12:22:37.000000 UTC N/A Disabled
3224 788 WinIPV6SE.exe 0xc9053dc6d320 13 - 0 False 2026-06-18 12:22:37.000000 UTC N/A Disabled
3552 660 SearchIndexer.exe 0xc9053dc63240 16 - 0 False 2026-06-18 12:22:38.000000 UTC N/A Disabled
3820 660 netcat.exe 0xc9053db24000 9 - 0 False 2026-06-18 12:22:39.000000 UTC N/A Disabled
2056 1544 AggregatorHost 0xc9053dfcb340 4 - 0 False 2026-06-18 12:22:40.000000 UTC N/A Disabled
4680 660 SecurityHealth 0xc9053e34f000 10 - 0 False 2026-06-18 12:22:47.000000 UTC N/A Disabled
4908 788 MoflsoCoreWorker 0xc9053e2f92e0 8 - 0 False 2026-06-18 12:22:53.000000 UTC N/A Disabled
5116 660 svchost.exe 0xc9053e550000 5 - 0 False 2026-06-18 12:22:58.000000 UTC N/A Disabled
4356 660 svchost.exe 0xc9053e2da000 8 - 0 False 2026-06-18 12:23:49.000000 UTC N/A Disabled
2696 660 svchost.exe 0xc9053da2d000 3 - 0 False 2026-06-18 12:23:49.000000 UTC N/A Disabled
4580 660 svchost.exe 0xc9053e0bd000 4 - 0 False 2026-06-18 12:24:41.000000 UTC N/A Disabled
3668 660 svchost.exe 0xc9053dfcf2e0 8 - 0 False 2026-06-18 12:24:41.000000 UTC N/A Disabled
2572 724 sihost.exe 0xc9053daf52e0 9 - 1 False 2026-06-18 12:25:36.000000 UTC N/A Disabled
3828 660 svchost.exe 0xc9053dcd2e0 9 - 1 False 2026-06-18 12:25:36.000000 UTC N/A Disabled
1328 724 taskhostv.exe 0xc9053e1b0000 6 - 1 False 2026-06-18 12:25:36.000000 UTC N/A Disabled
1960 376 ctfmon.exe 0xc9053d862000 11 - 1 False 2026-06-18 12:25:36.000000 UTC N/A Disabled
4144 592 explorer.exe 0xc9053e70a000 0 - 1 False 2026-06-18 12:25:36.000000 UTC 2026-06-18 12:26:06.000000 UTC Disabled
1120 4144 explorer.exe 0xc9053d095000 70 - 1 False 2026-06-18 12:25:37.000000 UTC N/A Disabled
720 660 svchost.exe 0xc9054030f000 7 - 1 False 2026-06-18 12:25:39.000000 UTC N/A Disabled
3536 788 StartMenuExperi 0xc905403c0000 7 - 1 False 2026-06-18 12:25:40.000000 UTC N/A Disabled
```

Activ
Go to

3536	788	StartMenuExper	0xc9b5403c8080	7	-	1	False	2026-06-18	12:25:40.000000 UTC	N/A	Disabled
4532	788	RuntimeBroker.	0xc9b5404e2080	3	-	1	False	2026-06-18	12:25:41.000000 UTC	N/A	Disabled
5228	788	SearchApp.exe	0xc9b53de60080	53	-	1	False	2026-06-18	12:25:42.000000 UTC	N/A	Disabled
5384	788	RuntimeBroker.	0xc9b540658080	11	-	1	False	2026-06-18	12:25:42.000000 UTC	N/A	Disabled
6464	788	RuntimeBroker.	0xc9b53de49080	5	-	1	False	2026-06-18	12:25:43.000000 UTC	N/A	Disabled
6440	1120	SecurityHealth	0xc9b540a840c0	1	-	1	False	2026-06-18	12:25:55.000000 UTC	N/A	Disabled
6556	1120	vntoolsd.exe	0xc9b53de7f080	4	-	1	False	2026-06-18	12:25:55.000000 UTC	N/A	Disabled
6600	1120	OneDrive.exe	0xc9b5406e0080	0	-	1	False	2026-06-18	12:25:56.000000 UTC	2026-06-18 14:50:34.000000 UTC	Disabled
6256	1120	nsedge.exe	0xc9b5410b3080	0	-	1	False	2026-06-18	12:25:57.000000 UTC	N/A	Disabled
7392	788	ApplicationPra	0xc9b540f38080	3	-	1	False	2026-06-18	12:26:53.000000 UTC	N/A	Disabled
1984	788	UserOOBhroker	0xc9b53e2c0080	5	-	1	False	2026-06-18	12:26:55.000000 UTC	N/A	Disabled
7172	660	svchost.exe	0xc9b54089e080	5	-	0	False	2026-06-18	12:27:39.000000 UTC	N/A	Disabled
1864	7236	nsedge.exe	0xc9b540f10080	0	-	1	False	2026-06-18	12:31:50.000000 UTC	2026-06-21 12:40:51.000000 UTC	Disabled
6556	788	SearchApp.exe	0xc9b540222080	12	-	1	False	2026-06-18	12:32:44.000000 UTC	N/A	Disabled
7636	6596	nsedge.exe	0xc9b5402be080	54	-	1	False	2026-06-18	12:32:45.000000 UTC	N/A	Disabled
5380	7636	nsedge.exe	0xc9b541fcb080	9	-	1	False	2026-06-18	12:32:45.000000 UTC	N/A	Disabled
8128	7636	nsedge.exe	0xc9b5428870c0	16	-	1	False	2026-06-18	12:32:47.000000 UTC	N/A	Disabled
7540	7636	nsedge.exe	0xc9b5411e8080	21	-	1	False	2026-06-18	12:32:47.000000 UTC	N/A	Disabled
6928	7636	nsedge.exe	0xc9b5413bd080	11	-	1	False	2026-06-18	12:32:47.000000 UTC	N/A	Disabled
6936	7636	nsedge.exe	0xc9b54145d080	19	-	1	False	2026-06-18	12:32:47.000000 UTC	N/A	Disabled
4720	788	TextInputHost.	0xc9b5408da080	10	-	1	False	2026-06-18	12:35:35.000000 UTC	N/A	Disabled
3528	788	dllhost.exe	0xc9b5411e6080	7	-	1	False	2026-06-18	12:35:35.000000 UTC	N/A	Disabled
7984	1120	nmc.exe	0xc9b540776080	18	-	1	False	2026-06-18	13:28:56.000000 UTC	N/A	Disabled
2484	4712	OneDrive.exe	0xc9b540721080	28	-	1	False	2026-06-18	14:00:37.000000 UTC	N/A	Disabled
7140	1864	nsedge.exe	0xc9b542913080	0	-	1	False	2026-06-21	12:40:51.000000 UTC	2026-06-21 12:43:33.000000 UTC	Disabled
10088	10064	MicrosoftEdgeU	0xc9b543fa3080	0	-	0	True	2026-06-21	12:41:14.000000 UTC	N/A	Disabled
9812	660	svchost.exe	0xc9b543fda080	2	-	0	False	2026-06-21	12:41:29.000000 UTC	N/A	Disabled
7144	8376	nsedge.exe	0xc9b544123340	0	-	1	False	2026-06-21	12:43:31.000000 UTC	2026-06-21 12:43:36.000000 UTC	Disabled
9844	8376	nsedge.exe	0xc9b542916080	0	-	1	False	2026-06-21	12:43:31.000000 UTC	2026-06-21 12:43:38.000000 UTC	Disabled
9860	8376	nsedge.exe	0xc9b541429080	0	-	1	False	2026-06-21	12:43:31.000000 UTC	2026-06-21 12:43:38.000000 UTC	Disabled
10080	8376	nsedge.exe	0xc9b542653080	0	-	1	False	2026-06-21	12:43:31.000000 UTC	2026-06-21 12:43:36.000000 UTC	Disabled
220	8376	nsedge.exe	0xc9b542655300	0	-	1	False	2026-06-21	12:43:34.000000 UTC	2026-06-21 12:43:38.000000 UTC	Disabled
5224	9884	nsedge.exe	0xc9b541121080	0	-	1	False	2026-06-21	12:43:36.000000 UTC	2026-06-21 12:44:12.000000 UTC	Disabled
9664	788	FileCohost.exe	0xc9b543da3080	6	-	1	False	2026-06-21	12:43:37.000000 UTC	N/A	Disabled
7688	9884	nsedge.exe	0xc9b543ddc080	0	-	1	False	2026-06-21	12:43:46.000000 UTC	2026-06-21 12:44:14.000000 UTC	Disabled
2632	9884	nsedge.exe	0xc9b5413ab080	0	-	1	False	2026-06-21	12:43:49.000000 UTC	2026-06-21 12:44:14.000000 UTC	Disabled
7268	9884	nsedge.exe	0xc9b5408b9080	9	-	1	False	2026-06-21	12:44:11.000000 UTC	2026-06-21 12:44:14.000000 UTC	Disabled
7240	8560	nsedge.exe	0xc9b5426ad080	0	-	1	False	2026-06-21	12:47:29.000000 UTC	2026-06-21 15:02:09.000000 UTC	Disabled
10132	1120	powershell.exe	0xc9b541129080	12	-	1	False	2026-06-21	12:51:50.000000 UTC	N/A	Disabled
2072	10132	conhost.exe	0xc9b54033c080	5	-	1	False	2026-06-21	12:51:50.000000 UTC	N/A	Disabled
8672	1120	powershell.exe	0xc9b540233080	9	-	1	False	2026-06-21	13:00:45.000000 UTC	N/A	Disabled
8496	8672	conhost.exe	0xc9b54042d080	5	-	1	False	2026-06-21	13:00:45.000000 UTC	N/A	Disabled
10040	1120	WireShark.exe	0xc9b540258080	9	-	1	False	2026-06-21	13:01:26.000000 UTC	N/A	Disabled
8492	788	ShellExperience	0xc9b540850080	14	-	1	False	2026-06-21	13:02:42.000000 UTC	N/A	Disabled
5640	788	RuntimeBroker.	0xc9b5404fb080	5	-	1	False	2026-06-21	13:02:43.000000 UTC	N/A	Disabled
3304	592	LogonUI.exe	0xc9b5426ed080	0	-	1	False	2026-06-21	13:49:28.000000 UTC	2026-06-21 14:22:04.000000 UTC	Disabled
7900	788	LockApp.exe	0xc9b541319080	12	-	1	False	2026-06-21	13:49:30.000000 UTC	N/A	Disabled
3848	788	RuntimeBroker.	0xc9b54309a080	2	-	1	False	2026-06-21	13:49:31.000000 UTC	N/A	Disabled
536	3860	OneDrive.Sync.	0xc9b53e1a8340	21	-	1	False	2026-06-21	14:29:02.000000 UTC	N/A	Disabled
5432	660	svchost.exe	0xc9b5443a7080	25	-	0	False	2026-06-21	14:57:32.000000 UTC	N/A	Disabled
9124	2572	M365Copilot.exe	0xc9b5414b7080	27	-	1	False	2026-06-21	14:59:15.000000 UTC	N/A	Disabled
8892	788	RuntimeBroker.	0xc9b543f98080	1	-	1	False	2026-06-21	14:59:17.000000 UTC	N/A	Disabled
10024	9124	nsedge.exe	0xc9b540845080	46	-	1	False	2026-06-21	14:59:18.000000 UTC	N/A	Disabled
9956	10024	nsedge.exe	0xc9b541113300	7	-	1	False	2026-06-21	14:59:18.000000 UTC	N/A	Disabled
10064	10024	nsedge.exe	0xc9b540a6d080	12	-	1	False	2026-06-21	14:59:20.000000 UTC	N/A	Disabled
9164	10024	nsedge.exe	0xc9b5438a5080	16	-	1	False	2026-06-21	14:59:20.000000 UTC	N/A	Disabled
8864	10024	nsedge.exe	0xc9b54156a080	9	-	1	False	2026-06-21	14:59:20.000000 UTC	N/A	Disabled
9232	10024	nsedge.exe	0xc9b544464080	16	-	1	False	2026-06-21	14:59:21.000000 UTC	N/A	Disabled
10076	8560	nsedge.exe	0xc9b5439130c0	52	-	1	False	2026-06-21	15:02:09.000000 UTC	N/A	Disabled
5424	10076	nsedge.exe	0xc9b5420c50c0	8	-	1	False	2026-06-21	15:02:09.000000 UTC	N/A	Disabled
4708	10076	nsedge.exe	0xc9b5406d70c0	17	-	1	False	2026-06-21	15:02:09.000000 UTC	N/A	Disabled
9608	10076	nsedge.exe	0xc9b54309b080	9	-	1	False	2026-06-21	15:02:09.000000 UTC	N/A	Disabled
7976	10076	nsedge.exe	0xc9b540c43080	10	-	1	False	2026-06-21	15:02:09.000000 UTC	N/A	Disabled
6644	1516	audiody.exe	0xc9b5414020c0	4	-	0	False	2026-06-21	15:04:13.000000 UTC	N/A	Disabled
4036	788	smactreen.exe	0xc9b5412e5080	1	-	1	False	2026-06-21	15:09:18.000000 UTC	N/A	Disabled
3296	660	svchost.exe	0xc9b541164080	3	-	0	False	2026-06-21	15:09:18.000000 UTC	N/A	Disabled
5004	8672	python.exe	0xc9b53deac080	1	-	1	False	2026-06-21	15:10:30.000000 UTC	N/A	Disabled

Q1: What suspicious processes were discovered?

A1: python.exe was found with PID 3004 which executed NulltackKatz.py, along with powershell.exe with PID 10132

Q2: What is the PID of python.exe?

A2: PID: 3004 — PPID: 8672 — Created at 2026-06-21 15:10:30

Q3: What is the PID of lsass.exe?

A3: PID: 684 — PPID: 524 (wininit.exe) — This was the target process that NulltackKatz attempted to access for credential dumping

Q4: What evidence confirms that monitoring was active during the attack?

A4: The presence of Sysmon.exe with PID 2196 and Wireshark.exe with PID 10040 confirms that the Blue Team was actively monitoring the system during the attack

Q5: What MITRE technique is associated with these findings?

A5: T1059.006 — Command and Scripting Interpreter: Python

Q1: What does windows.psscan reveal compared to windows.pslist?

A1: windows.psscan scans raw memory for process structures, revealing processes that may be hidden or terminated. It confirmed the presence of python.exe (PID: 3004) which executed NulltackKatz.py

Q2: Was Mimikatz found directly in memory?

A2: No direct Mimikatz process was found, because it was executed from within python.exe as a module (NulltackKatz.py calls Mimikatz internally)

Q3: What is the significance of DumpIt.exe appearing in memory?

A3: DumpIt.exe (PID: 8700, PPID: 10132) confirms that the RAM capture was performed deliberately as part of the forensic investigation by the Blue Team

Q4: What MITRE technique is associated with these findings?

A4: T1003.001 — OS Credential Dumping: LSASS Memory

4	System	0xc9053d01f080	0x504	Process	0x1478	lsass.exe	Pid 684
4	System	0xc9053d01f080	0x60c	Process	0x1028	lsass.exe	Pid 684
4	System	0xc9053d01f080	0x61c	Process	0x102a	lsass.exe	Pid 684
4	System	0xc9053d01f080	0x784	Process	0x102a	lsass.exe	Pid 684
4	System	0xc9053d01f080	0x78c	Process	0x102a	lsass.exe	Pid 684
4	System	0xc9053d01f080	0x790	Process	0x1fffff	lsass.exe	Pid 684
4	System	0xc9053d01f080	0x798	Process	0x102a	lsass.exe	Pid 684
444	csrss.exe	0xc9053d01f080	0x278	Process	0x1fffff	lsass.exe	Pid 684
684	lsass.exe	0xc9053d0038e0	0x4	Event	0x1f0003	-	-
684	lsass.exe	0xc9053d003ee0	0x8	Event	0x1f0003	-	-
684	lsass.exe	0xc9053d0032660	0xc	WaitCompletionPacket	0x1	-	-
684	lsass.exe	0xc9053d042c00	0x10	IoCompletion	0x1f0003	-	-
684	lsass.exe	0xc9053d01cd00	0x14	TpWorkerFactory	0xf00ff	-	-
684	lsass.exe	0xc9053933f4e0	0x18	IRTimer	0x100002	-	-
684	lsass.exe	0xc9053d0032180	0x1c	WaitCompletionPacket	0x1	-	-
684	lsass.exe	0xc90538f03090	0x20	IRTimer	0x100002	-	-
684	lsass.exe	0xc9053d0032590	0x24	WaitCompletionPacket	0x1	-	-
684	lsass.exe	0xc9053d041c20	0x28	EtwRegistration	0x804	-	-
684	lsass.exe	0xc9053d0410c0	0x2c	EtwRegistration	0x804	-	-
684	lsass.exe	0xc9053d0424e0	0x30	EtwRegistration	0x804	-	-
684	lsass.exe	0xda8908ed62a0	0x34	Directory	0x3	KnownDlls	-
684	lsass.exe	0xc9053d003960	0x38	Event	0x1f0003	-	-
684	lsass.exe	0xc9053d0039e0	0x3c	Event	0x1f0003	-	-
684	lsass.exe	0xc9053d02d6b0	0x40	File	0x100020	\Device\HarddiskVolume3\Windows\System32	-
684	lsass.exe	0xc9053d0411a0	0x44	EtwRegistration	0x804	-	-
684	lsass.exe	0xc905391c5a90	0x48	Mutant	0x1f0001	SM0:684:304:WilStaging_02	-
684	lsass.exe	0xc90539211d20	0x4c	ALPC Port	0x1f0001	-	-
684	lsass.exe	0xda89089bdce0	0x50	Directory	0xf	BaseNamedObjects	-
684	lsass.exe	0xc9053a1e9790	0x54	Semaphore	0x1f0003	SM0:684:304:WilStaging_02_p0	-
684	lsass.exe	0xc9053a1e9290	0x58	Semaphore	0x1f0003	SM0:684:304:WilStaging_02_p0h	-
684	lsass.exe	0xc9053d003ce0	0x5c	Event	0x1f0003	-	-
684	lsass.exe	0xc9053d0032f50	0x60	WaitCompletionPacket	0x1	-	-
684	lsass.exe	0xc9053d042cc0	0x64	EtwRegistration	0x804	-	-
684	lsass.exe	0xc9053d042da0	0x68	EtwRegistration	0x804	-	-
684	lsass.exe	0xc9053d042f60	0x6c	EtwRegistration	0x804	-	-
684	lsass.exe	0xc9053d042e80	0x70	IoCompletion	0x1f0003	-	-
684	lsass.exe	0xc9053981f9f0	0x74	TpWorkerFactory	0xf00ff	-	-
684	lsass.exe	0xc90538f02b40	0x78	IRTimer	0x100002	-	-
684	lsass.exe	0xc9053d00329a0	0x7c	WaitCompletionPacket	0x1	-	-
684	lsass.exe	0xc90538f03910	0x80	IRTimer	0x100002	-	-
684	lsass.exe	0xc9053d0032b40	0x84	WaitCompletionPacket	0x1	-	-

Q1: What processes were found accessing lsass.exe?

A1: System (PID 4) and csrss.exe (PID 444) were found with handles to lsass.exe (PID 684)

Q2: What is the significance of access mask 0x1fffff?

A2: 0x1fffff represents FULL ACCESS to lsass.exe, which is the highest possible access level and a strong indicator of credential dumping (T1003.001)

Q3: What MITRE technique does this confirm?

A3: T1003.001 — OS Credential Dumping: LSASS Memory

System Information — windows.info

The memory image was successfully analyzed using Volatility 3. The dump belongs to a 64-bit Windows 10 machine (Version 10.0.19041) captured on 2026-06-21 at 15:13:10 UTC using DumpIt v3.6. The system runs on 2 processors and the memory layer was identified as WindowsCrashDump64Layer, confirming the integrity of the forensic image

Process Analysis — windows.pslist & windows.psscan

Process analysis revealed several critical findings. python.exe (PID: 3004, PPID: 8672) was identified as the malicious process responsible for executing NulltackKatz.py at 15:10:30 UTC. powershell.exe (PID: 10132) was also present, confirming it served as the parent process that launched the attack. lsass.exe (PID: 684) was running under its legitimate parent wininit.exe (PID: 524), confirming it was the target for credential dumping. No direct Mimikatz process was found because it was executed internally through python.exe. DumpIt.exe (PID: 8700) confirmed the deliberate RAM capture performed by the Blue Team.

LSASS Handle Access — windows.handles

Handle analysis confirmed that System (PID: 4) and csrss.exe (PID: 444) had active handles to lsass.exe (PID: 684) with an access mask of 0x1fffff, representing FULL ACCESS — the highest possible privilege level. This is the strongest memory-based indicator of credential dumping activity targeting LSASS, directly supporting the Sysmon Event ID 10 findings captured during the live attack phase

What is the PID of Mimikatz in memory?

A1: Mimikatz did not appear as a standalone process in memory. It was executed internally within python.exe (PID: 3004) through the NulltackKatz.py script, which called Mimikatz as a module. This is a common evasion technique to avoid direct detection in process listings.

Does the memory dump show the executed commands?

A2: The memory dump did not explicitly show the executed commands in plaintext. However, the forensic evidence collected confirms the attack chain:

- python.exe (PID: 3004) executed NulltackKatz.py
- lsass.exe (PID: 684) was accessed with full privileges (0x1fffff)
- Sysmon Event ID 10 logged the LSASS memory access during the live attack

This indirect evidence strongly supports that Mimikatz commands were executed through the Python script to dump credentials from LSASS memory — consistent with **T1003.001**

Group Identity & Attribution

Alias / Designation	Assigned By	Context / Notes
APT28	Mandiant / FireEye	Advanced Persistent Threat designation
Fancy Bear	CrowdStrike	Most widely-known public name
Sofacy / SOFACY	Kaspersky Lab	Named after malware family
STRONTIUM	Microsoft (MSTIC)	Used in Microsoft legal filings
Pawn Storm	Trend Micro	Named after chess strategy
BlueDelta	Recorded Future / Insikt Group	Used in 2024–2025 campaigns
Sednit	ESET	Named after Sednit malware toolkit
Iron Twilight	SecureWorks CTU	SecureWorks threat group tracking name
Group 74	Talos Intelligence	Cisco Talos internal designation
TG-4127	Dell SecureWorks	Early threat-group identifier
TA422	Proofpoint	Email-threat actor designation
Forest Blizzard	Microsoft (2023 rebrand)	Latest Microsoft naming taxonomy

Historical Operations

Operation	Year	Target	Techniques Used
U.S. Election Interference	2016	DNC, Hillary Clinton campaign, DCCC	Spearphishing, credential harvesting, X-Agent malware, LSASS dumping
WADA / USADA Operations	2014–2018	Anti-doping agencies, Olympic athletes	Spearphishing, web exploitation, data theft and leak operations
OPCW / Spiez Laboratory Attacks	2018	Chemical weapons investigators	Wi-Fi proximity attacks, malicious USB, credential theft
BlueDelta Credential Harvesting	2025	Turkish energy sector, European think tanks	Spearphishing, credential harvesting, TLS exfiltration via Gmail
French Election Interference (Macron)	2017	En Marche! campaign, French political parties	Spearphishing, document theft, influence operations via social media

Technical TTP Analysis

Phase	MITRE ID	Technique	Tactic	Implemented in
1	T1566.001	Spearphishing Attachment	Initial Access	Phase1_spearphishing()
2	T1204.002	Malicious File Execution	Execution	Phase2__execution
3	T1087.001	Local Account Discovery	Discovery	Phase3_discovery()-net user
4	T1016	Network Configuration	Discovery	Phase3_discovery()-ipconfig/all
5	T1057	Process Discovery	Discovery	Phase3_discovery()-tasklist
6	T1134	Access Token Manipulation	Privilege Escalation	Phase4_privilegeescalation
7	T1003.001	LSASS memory Dumping	Credential Access	Phase5_credentialdump()

Infrastructure & Indicators of Compromise (IoCs)

Domains Used by APT28 (with Purpose)

APT28 has registered and used attack-delivery domains such as **wellnessmedcare[.]org**, **wellnesscaredmed[.]com**, **freefoodaid[.]com**, and **longsauce[.]com** to host and deliver malicious Office exploit documents and LNK files — with threat actors sometimes registering new domains the same day as attacks, reflecting highly agile operations.

[Trellix](#)

A fifth notable domain comes from credential-harvesting operations: **accesscam[.]org** — registered to dynamic DNS provider Dynu Systems — was identified as APT28 infrastructure by both CISA and Google's Threat Intelligence Group, used as part of their credential theft campaigns targeting Western logistics and technology entities

Domain	Purpose
Wellnessmedcare(.)org	Delivery of malicious Office exploit documents (2026 campaign)
Wellnesscaredmed(.)com	Hosting LNK files and exploit payloads for spear-phishing
Freefoodaid(.)com	Staging server for multi-stage malware chain
Longsauce(.)com	Decoy content delivery and WebDAV exploit hosting
Accesscam(.)org	Credential harvesting / phishing infrastructure (Western logistics campaign)

IP Addresses Attributed to APT28

IP Address	Context
185.176.43[.]106	APT28 C2 server (Cisco router campaign)
5.199.174[.]219	APT28 malicious infrastructure

162.241.216[.]236	C2 server used by CredoMap stealer (data exfiltration via IMAP port 143)
158.58.173[.]40	APT28 attributed C2 node
192.145.125[.]42	APT28 attributed C2 node

Malware / Context	Type	Hash
winexesvc.exe — Winexe remote execution tool used in the German Bundestag compromise (Artifact #1, retrieved from a Die Linke file server)	SHA 256	5130f600cd9a9cdc82d4bad938b20cbd2f699aadb76e7f3f1a93602330d9997d
svchost.exe.exe — Second artifact from the Bundestag breach; compiled 2015-04-22	SHA 256	730a0e3daf0b54f065bdd2ca427fbe10e8d4e28646a5dc40cbcfb15e1702ed9a
CredoMap — A .NET credential stealer targeting Chrome, Firefox, and Edge, exfiltrating data via IMAP to a hardcoded C2; used against Ukrainian targets exploiting CVE-2022-30190 (Follina)	SHA 256	2318ae5d7c23bf186b88abecf892e23ce199381b22c8eb216ad1616ee8877933
BeardShell loader — APT28 implant reported by Sekoia, used in European government and military targeting; injected into explorer.exe via steganography	SHA 256	88e28107fbf171fdbcf4abbc0c731295549923e82ce19d5b6f6fefa3c9f497c9
СЛУЖБОВА ХАРАКТЕРИСТ	MD5	608877a9e11101da53bce99b0effc75b

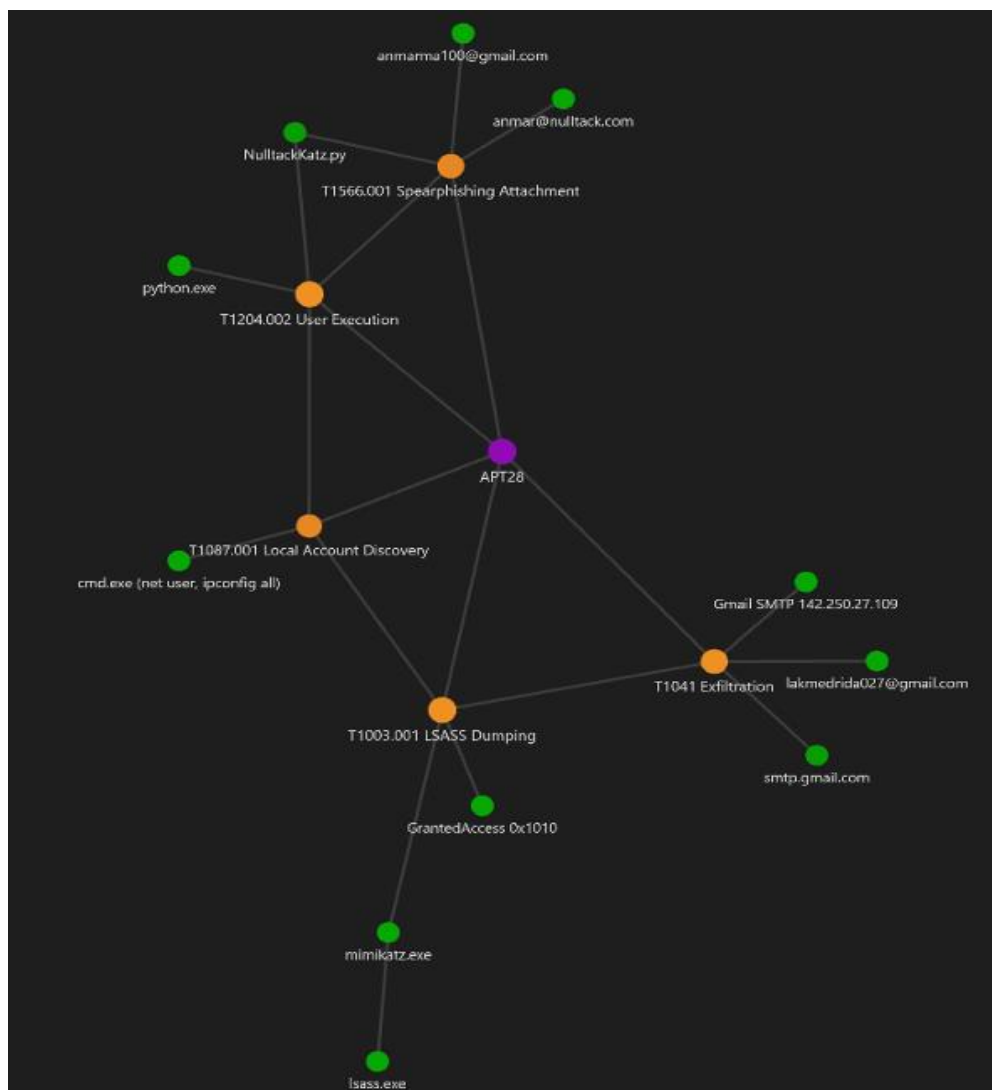
IKA.doc — A malicious Word document with embedded VBA macros used in Operation Phantom Net Voxel; drops a DLL and PNG shellcode loader upon execution		
--	--	--

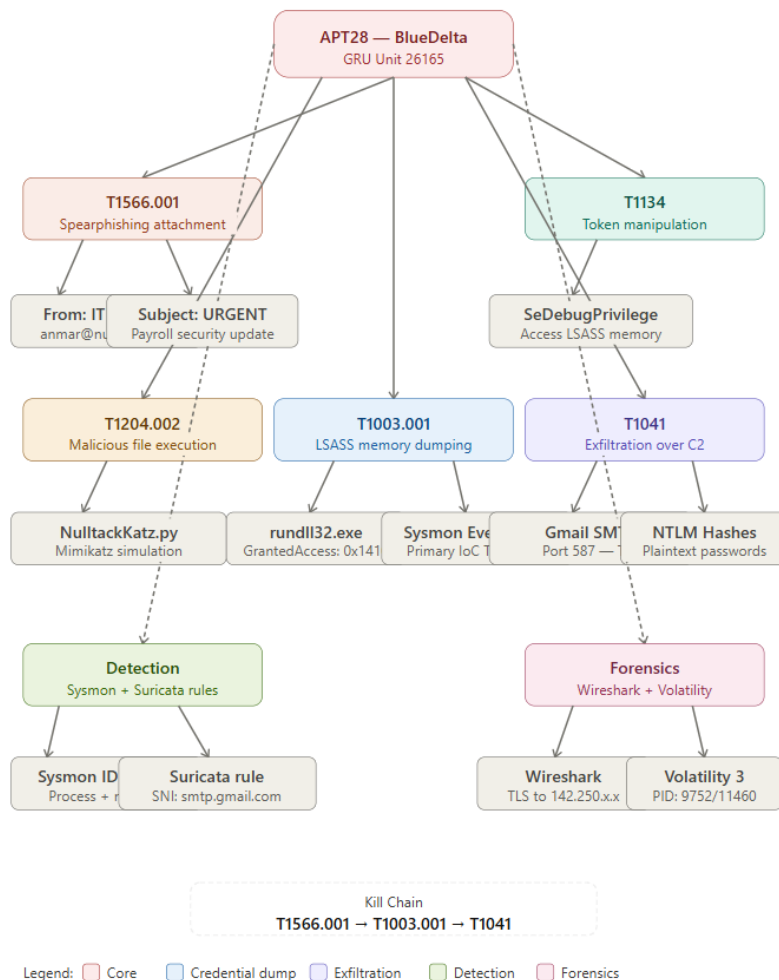
Key Takeaways

- APT28 rapidly rotates infrastructure — sometimes registering domains the same day they are used in attacks, making static IoC blocking insufficient on its own
- APT28 leverages legitimate cloud storage services such as Icedrive and Koofr for C2 operations, using them as dead-drop resolvers to host malware payloads and store exfiltrated data — blending into normal traffic
- Since 2024 and into 2026, APT28 has been configuring Virtual Private Servers (VPSs) to operate as malicious DNS servers, exploiting small office/home office (SOHO) routers to redirect victim traffic through attacker-controlled infrastructure

Type	Recommendation	Threat Addressed
Detection	Monitor unusual IMAP traffic	CredoMap / Data Exfiltration
Detection	Monitor Macro & DLL Drops	DNS Hijacking / AitM Attacks
Detection	Monitor Macro & DLL Drops	Initial Access via Office docs
Architectural	Zero Trust Architecture	Lateral Movement post-breach
Architectural	Management Interface Isolation	SNMP / RDP Exploitation
Architectural	DNS Security + Egress Filtering	Cloud-based C2 / DNS Hijacking

Knowledge Graph&attack Workflow





Lessons Learned

One of the most important lessons I learned from this research is to never open any message or attachment without first verifying the identity of the sender personally. Due to the increasing number of cyberattacks and hacking operations carried out by groups like APT28, I now understand that even messages that appear to come from trusted sources can be weaponized. Attackers often use compromised real accounts to make their phishing emails look completely legitimate, which makes it nearly impossible to detect the threat just by looking at the sender's name. Therefore, before opening any suspicious message, I must confirm directly with the sender through a separate and trusted communication channel, whether by phone or in person, to ensure the message is genuinely from them.

Conclusion

APT28 is one of the most dangerous state-sponsored threat actors in the world, linked to Russia's GRU military intelligence. The group uses advanced malware, rotating infrastructure, and sophisticated techniques such as spear-phishing, DNS hijacking, and cloud-based C2 channels to target governments and critical infrastructure globally. What makes APT28 especially dangerous is its ability to adapt quickly and weaponize new vulnerabilities within hours of their release. Defending against this threat requires Zero Trust architecture, DNS filtering, and most importantly, human awareness — because the most effective entry point is often a single employee who opens the wrong email.